Affine Pooled Quartics in 215 Integers

Research iteration 22

26 October 2026

1 Affine pooled quartics in 215 unrestricted integers

Literal results. The two fully collected integer polynomials supplied with this section are

$$P_{215}(\boldsymbol{x}) = \sum_{i=0}^{151} R_i(\boldsymbol{x})^2, \qquad \boldsymbol{x} \in \mathbb{Z}^{215}.$$

The complete quadratic residuals, all collected quartic coefficients, and every coordinate map are included in the accompanying packet. The two residual bases give the following measured tradeoff:

Profile	Variables	Rows	Degree	Terms	Height
Sparse principal	215	152	4	14 578	16 866
Low-height alternative	215	152	4	34 700	4 512

Each residual coefficient has magnitude at most 64. All coordinates occur. The sparse form improves the prior 217-variable sparse equation in dimension, rows, and support at unchanged height. It also improves the incoming Pooled Projection 216-variable point, with 15 158 terms and height 334 952, in all four measures. The low-height form trades additional support for smaller coefficients. No global optimum is claimed.

Theorem and scope. For either literal polynomial, the ordinary proof below, composed with the retained numerical C222/ZF/ZFC refutation calibration, establishes in ZFC

$$\neg\exists \boldsymbol{x} \in \mathbb{Z}^{215}\ P_{215}(\boldsymbol{x}) = 0 \quad \longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This does not assert either side separately. The construction is direct through the numerical proof calculus, not K215 simulation or Jones specialization. The named logical interfaces are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. No new end-to-end proof-assistant compilation is claimed.

Attribution and new compatibility step. The immediate input is the retained Round21 `beta217` equation. Source 160 already proves the affine-horizon, weak affine-frame, difference-error, and small-horizon estimates used here. Source 158 supplies fixed-base pooling and the shared-private-consumer mechanism; source 159 supplies the inherited radical projections. The new compatibility argument strengthens the Pell frame enough to make fixed-base pools work with absolute atom bounds instead of squared-atom bounds. The resulting composition saves two further coordinates and rows.

Weak atom bounds with a fourfold Pell frame

Set

$$t = 8h, \qquad d = t + 2, \qquad F = d^t, \qquad q = 4z + 1, \qquad y = F(2v + 1).$$

The powers here describe meanings of retained ordinary polynomial certificates, not primitive operations in the final equation. Change the old growth request from base t to base d, retaining its index t; replace the old square-horizon condition by the affine parameterization.

The 35 affine atoms consisting of the 34 retained actual atoms and the new base d use 43 pre-Pell coordinates u_j. They cover all bases, indices and outputs of all 17 requests. For integers a, u,

$$|au| \le u^2 + \lfloor a^2/4 \rfloor.$$

Consequently $L = c + \sum a_j u_j$ satisfies $|L| \le \sum_{a_j \neq 0} u_j^2 + |c| + \sum \lfloor a_j^2/4 \rfloor$. The exact maximum overhead is 21, attained by the affine derived-label expression; the new growth base has overhead 18. Thus the single row

$$W = 22 + \sum_{j=1}^{43} u_j^2 \tag{AF}$$

gives $W \geq 22$ and $|L| < W$ for every actual request atom. The full atom/port ledger is checked independently. No private Pell coordinate occurs in it. Importantly, this does *not* assert $L^2 < W$.

Replace the two old frame multiplication rows by

$$U = 4W(2z_f - 1), \qquad V = (4W+1)U,$$

and retain $A^2 + U^2 - V^2 = 1$. Pell classification at center $4W+1$ and $Y_\ell(4W+1) \equiv \ell \pmod{4W}$ force a nonzero index of magnitude at least $4W$. Hence

$$|A| > (4W)^{4W}, \qquad |A-1| > 4W^3. \tag{PG}$$

For completeness, the choice $\ell = 4W$ has $Y_{4W}(4W+1)/(4W)$ odd and supplies an integral z_f. The finite arithmetic front determines W before this private Pell choice, so there is no circular frame dependence.

Retain the entire signed semantic-anchor cap and every principal-pair norm. In the orientation pool replace only

$$(Y_{F+t} - F - t)^2 \quad \text{by} \quad (Y_{F+t} - Y_t - F)^2,$$

retaining $(Y_t - t)^2$ and all abscissa errors. The two error systems have the same zero set. As in source 160, first exclude the exceptional unit anchor ordinate by the cap and its nonzero orientation residue; only then apply the comparison-index theorem. Classified indices have magnitude below W. The reduced nonnegative orientation error satisfies

$$0 \leq E_{\mathrm{or}} < 65W^2 + 60 < 4W^3 < |A-1|.$$

Its divisibility by $A-1$ forces every reduced original orientation error to vanish. Thus canonical signed Pell pairs are available before any arithmetic sign is assumed. This argument applies to either sign of A.

Pooling with absolute bounds

Let $r \leq 15$, let $n \neq 0$, and suppose $|n|, |m_i|, |k_i| < W$, with canonical signed Pell pairs X_i, Y_i at A and retained gates $Z_i = AY_i$. Write

$$\begin{aligned} T &= 2An - n^2 - 1, \\ H &= \sum_i Y_i(m_i - X_i), \\ S &= \sum_i \big((m_i - X_i + Z_i)^2 - Y_i^2\big). \end{aligned}$$

Use only the two existing pooled coordinates K, L and quadratic rows

$$K - 2H - (2A-n)L = 0, \qquad S - nK + L = 0. \tag{PP}$$

The exact polynomial identity modulo the retained AY gates is

$$\sum_i (m_i - X_i + Z_i - nY_i)^2 = S - 2nH - T\sum_i Y_i^2.$$

Thus T divides the squared-error sum. Since $T \equiv -1 \pmod{n}$, negative indices can be cleared modulo T. Put $b = W - 1$. Then T divides

$$E = \sum_i \big(m_i n^b - n^{b+k_i}\big)^2 \geq 0,$$

and every exponent displayed is nonnegative. The elementary bounds are

$$E \leq 60b^{4b} < (4W)^{4W}, \qquad |T| \geq 2|A| - (W-1)^2 - 1 > |A| > (4W)^{4W}.$$

Hence $E = 0$. At $|n| \geq 2$ this gives $k_i \geq 0$ and $m_i = n^{k_i}$; a negative index would imply $m_i n^{-k_i} = 1$. At $n = \pm 1$ it gives the exact signed unit value without asserting a nonnegative index. The $r = 1$ argument also revalidates all three singleton requests under the new weak frame. The converse uses the exact quotients $L = (S - 2nH)/T$ and $K = 2H + (2A-n)L$.

The two blocks have eleven requests at $q+1$ and three at q. This is a new size-bound composition of the credited pooling rows, not a new claim to those rows themselves.

Horizon, signs, and the inherited arithmetic budgets

For every integer h, the base $d = 8h + 2$ is nonzero and nonunit. Its canonical singleton certificate forces $t \geq 0$ and $F = d^t$. If $t = 0$, then $F = 1$ and y is odd. The endpoints read

$$x = 2 + (y + 1)u_0, \qquad x = 1 + (y + 1)u_t,$$

so $1 = (y + 1)(u_t - u_0)$ would be even. Therefore $t \geq 8$ and $F \geq 10^8$.

The retained signed bootstrap is used in its noncircular order. Since $q = 4z + 1$, first $|q + 1| \geq 2$, so the $q + 1$ pool gives nonnegative actual representatives and the nonzero semantic-anchor label. Hence $H \geq 2$; the row $B\mu = C + F + 1$ gives $B \neq 0$; the singleton at BH gives $x \geq 0$. Because y is a nonzero multiple of F, the last endpoint excludes $x = 0$. Thus $|q| \geq 2$, and only now the q pool gives $y + 1 \geq 0$, whence $y \geq F > 0$. The endpoints yield

$$x \geq F + 1, \qquad x \geq 2 + (t + 1)y \geq 2 + (t + 1)F.$$

Short signed-binomial recovery subsequently gives $B > 0$ and $q > 0$.

At the new smallest horizon one must not reuse $x > 2^{32}$. Instead $x \geq 900000002 > 2^{29}$; for $E_0 = 2^{23}$ and $2^s \leq x < 2^{s+1}$, $s \geq 29$,

$$E_0 \log_2(2x) < 2^{23}(s + 2) < 2^s \leq x.$$

Thus the retained local-polynomial bound still lies below $2^x < N$. Also $F > 2^{22}$ supplies the derived-label budget. The selector inequality holds directly at $t = 8$ because $840^2 < 10^8$; at every other positive multiple of eight, $\Lambda_t^2 \leq 16^{t-1} < (t + 2)^t = F$. The strict local ranges, all-primes CRT coverage and positive Horn-flow extraction therefore remain valid. Label multiplicities are recovered at the still-larger evaluation base $F > t^t$. The inherited $y = F$ boundary is excluded after local recovery, not prematurely.

Conversely choose a cofinally large positive multiple of eight t beyond all finite proof parameters. Then $v_2(F) = t > v_2(t!)$; the inherited CRT construction supplies arbitrarily large admissible odd-square beta multipliers, which are allowed by the retained odd-linear condition. Choose even x and the parent's parity-prescribed representatives. The binomial B and H are odd, so $q = (BH)^x \equiv 1 \pmod 4$. After constructing all arithmetic wires, choose the weak frame and strong Pell frame, canonical principal witnesses, and the pooled quotients. This is a fresh completeness construction, not a map preserving the old square horizon or growth value.

An additional exact shared-consumer contraction

In the pinned 217-coordinate chart put

$$F = x_3,\ S = x_{12},\ S' = x_{13},\ a = x_8,\ b = x_9,\ s = x_{19},\ K = x_{62},$$

and $p = x_{75}, r = x_{76}, g = x_{77}, v = x_{78}, e_1 = x_{79}, e_2 = x_{80}$. The old rows include

$$r_K = K - FS', \quad r_p = p - 1 - Fb, \quad r_r = r - 1 - F(a - 1), \quad r_g = g - 1 + 2F + 2Fa + 4Fb - 2Fs.$$

Delete p, r and their defining rows, and introduce only $J_{\text{prod}} = FS$. Replace the three consumers by

$$\begin{aligned} C_{61} &= -S' - Kb + gv + e_1, \\ C_{64} &= -S - J_{\text{prod}}(a - 1) + v, \\ C_{65} &= S + J_{\text{prod}}b - S' - K(2a + 4b + 2s) + e_2. \end{aligned}$$

Writing $r_J = J_{\text{prod}} - FS$ and $D = 2a + 4b + 2s$, the exact identities are

$$\begin{aligned} C_{61} &= R_{61} + S'r_p - br_K, \\ C_{64} &= R_{64} + Sr_r - (a - 1)r_J, \\ C_{65} &= R_{65} + br_J + (8S' - S)r_p + 4S'r_r + S'r_g - Dr_K. \end{aligned}$$

The forward map adjoins $J_{\text{prod}} = FS$ and drops p, r; the inverse restores $p = 1 + Fb$ and $r = 1 + F(a - 1)$. No other row uses the deleted coordinates. These maps are mutually inverse on integer zero sets. This is a specialized application of source 158's consumer method after earlier contractions, not a second credit for the already removed joint-square or third factor. Together with the affine horizon it changes 217/154 to 215/152.

Finite certification. The packet retains the complete parent archive, explicit source edits, all forward and backward maps, literal coefficients and frozen row operations. A separate checker uses only the standard library, imports no producer, and verifies complete ideal identities, all 17 actual request ports, the 43-coordinate major-ant, every inverse row operation, all collected coefficients and coordinate occurrence. Corrupted frame, growth and restoration cases must fail. Both normal and optimized Python, read-only reconstruction and retained parent replay are supplied. These finite checks do not replace the unbounded proof or decide consistency.

A Complete sparse residuals

$$R_0 = 8x_0 - 8x_0x_2 - x_2 + x_7$$

$$R_1 = -x_2x_5 - x_3 + x_6x_7$$

$$R_2 = -1 + x_1 - x_6x_7$$

$$R_3 = 1 + x_3 - x_4x_{18} + x_{16}$$

$$R_4 = -1 - x_8x_8 - 2x_8x_9 - x_9x_9 + x_{19}$$

$$\begin{aligned} R_5 = &-10x_8 + x_8x_8 - 6x_8x_9 - 8x_8x_{19} - 20x_9 \\ &- 11x_9x_9 - 16x_9x_{19} - 15x_{19} - 4x_{19}x_{19} + x_{20} \end{aligned}$$

$$\begin{aligned} R_6 = &-2 + 2x_8 - x_8x_8 - 4x_8x_{20} - 4x_{20} \\ &- 4x_{20}x_{20} + x_{21} \end{aligned}$$

$$\begin{aligned} R_7 = &1 + 6x_8 - 36x_8x_8 - 24x_8x_{21} + 2x_{21} \\ &- 4x_{21}x_{21} + x_{22} \end{aligned}$$

$$\begin{aligned} R_8 = &-2 - x_9 - x_9x_9 - 2x_9x_{10} - x_{10} \\ &- x_{10}x_{10} + x_{23} \end{aligned}$$

$$\begin{aligned} R_9 = &-1 + x_8 - x_8x_8 - 2x_8x_{10} + x_{10} \\ &- x_{10}x_{10} + x_{24} \end{aligned}$$

$$\begin{aligned} R_{10} = &-9 + 28x_8 + 8x_8x_{10} - 16x_8x_{19} - 8x_8x_{24} \\ &+ 26x_9 + 8x_9x_9 - 22x_9x_{10} + 16x_9x_{23} - 12x_9x_{24} \\ &+ 31x_{10} + 27x_{10}x_{10} - 24x_{10}x_{19} + 24x_{10}x_{23} - 8x_{19}x_{24} \\ &- x_{20} + 33x_{23} + 4x_{23}x_{23} + x_{25} - x_{29} \\ &- x_{43} \end{aligned}$$

$$\begin{aligned} R_{11} = &-16 + 14x_8 - 2x_8x_8 + 8x_8x_{10} - 8x_8x_{23} \\ &- 8x_{10} + 8x_{23} - 4x_{25}x_{25} + x_{26} - 2x_{43} \end{aligned}$$

$$\begin{aligned} R_{12} = &-2 - 8x_8 + 8x_8x_{23} + 8x_8x_{24} - 8x_9 \\ &+ 8x_9x_{23} + 8x_9x_{24} - 4x_{19} + 8x_{23} - 4x_{23}x_{23} \\ &- 8x_{23}x_{24} + 8x_{24} - 4x_{24}x_{24} + x_{27} \end{aligned}$$

$$\begin{aligned} R_{13} = &6 + 4x_8 + 32x_8x_9 + 16x_8x_{19} + 16x_8x_{24} \\ &+ 8x_8x_{27} + 24x_9 - 32x_9x_{24} - 16x_9x_{27} + 12x_{19} \\ &- 16x_{19}x_{24} - 8x_{19}x_{27} - x_{20} + 8x_{24} - 16x_{24}x_{24} \\ &- 16x_{24}x_{27} + 4x_{27} - 4x_{27}x_{27} + x_{28} \end{aligned}$$

$$\begin{aligned} R_{14} = &41 - 30x_8 - 8x_8x_8 - 16x_8x_9 - 42x_8x_{10} \\ &- 4x_8x_{23} - 16x_8x_{24} - 5x_9 + 5x_9x_9 - 16x_9x_{10} \\ &- 16x_9x_{23} - 4x_9x_{24} - 47x_{10} - 41x_{10}x_{10} - 24x_{10}x_{23} \\ &- 24x_{10}x_{24} - 34x_{23} - 4x_{23}x_{23} - 17x_{24} - 4x_{24}x_{24} \\ &+ x_{29} \end{aligned}$$

$$\begin{aligned} R_{15} = &46 - 10x_8 - 2x_8x_8 - 16x_8x_9 - 24x_8x_{10} \\ &- 8x_8x_{23} - 16x_9 - 32x_9x_{10} - 32x_9x_{23} - 40x_{10} \end{aligned}$$

$$
\begin{aligned}
&- 40x_{10}x_{10} - 48x_{10}x_{23} - 56x_{23} - 8x_{23}x_{23} - 4x_{29}x_{29} \\
&+ x_{30}
\end{aligned}
$$

$$
\begin{aligned}
R_{16} = \; &- 19x_8 + 8x_8x_8 + 8x_8x_{19} + 8x_8x_{23} - 46x_9 \\
&- 24x_9x_9 + 24x_9x_{19} + 24x_9x_{23} - 2x_{10} + 4x_{19} \\
&- 4x_{19}x_{19} - 8x_{19}x_{23} + 14x_{23} - 4x_{23}x_{23} - x_{24} \\
&+ x_{31}
\end{aligned}
$$

$$
\begin{aligned}
R_{17} = \; &- 3 - 6x_8 - 4x_{10} - 2x_{24} - 4x_{31} \\
&- 4x_{31}x_{31} + x_{32}
\end{aligned}
$$

$$
\begin{aligned}
R_{18} = \; &8 + 42x_8 + 7x_8x_8 + 28x_8x_{20} + 48x_9 \\
&- 32x_9x_9 - 32x_9x_{19} + 64x_9x_{20} - 40x_{19}x_{19} + 32x_{19}x_{20} \\
&+ 14x_{20} - 4x_{20}x_{20} + x_{33} + 2x_{36}
\end{aligned}
$$

$$
\begin{aligned}
R_{19} = \; &1 + 6x_8 - 36x_8x_8 - 24x_8x_{33} + 2x_{33} \\
&- 4x_{33}x_{33} + x_{34}
\end{aligned}
$$

$$
R_{20} = \; - x_9x_9 - 4x_9x_{34} - 4x_{34}x_{34} + x_{35}
$$

$$
\begin{aligned}
R_{21} = \; &- 6 - 2x_8 - 4x_9 - 14x_{19} - 16x_{19}x_{19} \\
&+ x_{36}
\end{aligned}
$$

$$
\begin{aligned}
R_{22} = \; &24 + 21x_9 - 21x_9x_9 - 18x_9x_{10} - 24x_9x_{23} \\
&- 24x_9x_{36} + 3x_{10} - 9x_{10}x_{10} - 16x_{10}x_{36} + 3x_{23} \\
&- 12x_{23}x_{23} - 8x_{23}x_{36} + 4x_{36} - 4x_{36}x_{36} + x_{37} \\
&+ 2x_{43}
\end{aligned}
$$

$$
\begin{aligned}
R_{23} = \; &16 - 10x_8 - 20x_8x_8 - 16x_8x_{10} - 24x_8x_{24} \\
&- 24x_8x_{37} - 12x_{10} - 16x_{10}x_{24} - 16x_{10}x_{37} - 14x_{24} \\
&- 4x_{24}x_{24} - 8x_{24}x_{37} + 4x_{37} - 4x_{37}x_{37} + x_{38}
\end{aligned}
$$

$$
R_{24} = \; - x_{10}x_{10} - 2x_{10}x_{11} - x_{11}x_{11} + x_{39}
$$

$$
\begin{aligned}
R_{25} = \; &- 29 + x_8x_8 - 4x_8x_{39} - 14x_{10} - 2x_{10}x_{10} \\
&- 16x_{10}x_{23} + 16x_{10}x_{24} - 8x_{10}x_{39} - 8x_{23} + 12x_{23}x_{23} \\
&- 8x_{23}x_{24} + 22x_{24} - 4x_{24}x_{24} + 4x_{39} - 4x_{39}x_{39} \\
&+ x_{40} - 4x_{43}
\end{aligned}
$$

$$
\begin{aligned}
R_{26} = \; &4 + 2x_8x_8 - 8x_8x_{39} - 4x_{10}x_{10} - 16x_{10}x_{39} \\
&- 4x_{24} - 16x_{24}x_{24} - 32x_{24}x_{39} - 16x_{24}x_{40} + 4x_{39} \\
&- 24x_{39}x_{39} - 16x_{39}x_{40} - 4x_{40}x_{40} + x_{41}
\end{aligned}
$$

$$
\begin{aligned}
R_{27} = \; &- 16x_8x_8 + 8x_8x_{10} + 32x_8x_{24} + 16x_8x_{27} + x_{10} \\
&- x_{10}x_{10} - 8x_{10}x_{24} - 4x_{10}x_{27} - 16x_{24}x_{24} - 16x_{24}x_{27} \\
&- 4x_{27}x_{27} + x_{42}
\end{aligned}
$$

$$
\begin{aligned}
R_{28} = \; &- 1 - 8x_{10} - 4x_{10}x_{10} + 8x_{10}x_{23} + 8x_{23} \\
&- 4x_{23}x_{23} + x_{43}
\end{aligned}
$$

$$
\begin{aligned}
R_{29} = \; &- 1 - 9x_{10}x_{10} + 24x_{10}x_{23} + 12x_{10}x_{43} - 16x_{23}x_{23} \\
&- 16x_{23}x_{43} - 4x_{43}x_{43} + x_{44}
\end{aligned}
$$

$$R_{30} = 1 + x_9 - x_9x_9 - 4x_9x_{10} - 4x_9x_{44} - 4x_{10}x_{10} - 8x_{10}x_{44} - 4x_{44}x_{44} + x_{45}$$

$$R_{31} = -1 - 4x_{45} - 4x_{45}x_{45} + x_{46}$$

$$R_{32} = -1 - x_{10}x_{10} - 8x_{10}x_{45} - 4x_{10}x_{46} - 16x_{45}x_{45} - 16x_{45}x_{46} - 4x_{46}x_{46} + x_{47}$$

$$R_{33} = 2 + x_9 - x_9x_9 - 4x_9x_{10} - 4x_9x_{47} - 4x_{10}x_{10} - 8x_{10}x_{47} - 4x_{47}x_{47} + x_{48}$$

$$R_{34} = -4x_{42}x_{42} - 8x_{42}x_{48} - 4x_{48}x_{48} + x_{49}$$

$$R_{35} = -9 + 7x_8 - x_8x_8 - 8x_8x_{42} - 4x_8x_{49} + 24x_{42} - 16x_{42}x_{42} - 16x_{42}x_{49} + 12x_{49} - 4x_{49}x_{49} + x_{50}$$

$$R_{36} = -1 - 4x_{50} - 4x_{50}x_{50} + x_{51}$$

$$R_{37} = -x_{10}x_{10} - 8x_{10}x_{50} - 4x_{10}x_{51} - 16x_{50}x_{50} - 16x_{50}x_{51} - 4x_{51}x_{51} + x_{52}$$

$$R_{38} = -x_8 - x_8x_8 - 4x_8x_{10} - 4x_8x_{52} - 4x_{10} - 4x_{10}x_{10} - 8x_{10}x_{52} - 4x_{52} - 4x_{52}x_{52} + x_{53}$$

$$R_{39} = -10 - 12x_{53} - 4x_{53}x_{53} + x_{54}$$

$$R_{40} = 2 + 4x_{10} + 16x_{10}x_{53} + 8x_{10}x_{54} - 4x_{23} - 16x_{23}x_{53} - 8x_{23}x_{54} - x_{43} + 8x_{53} - 16x_{53}x_{53} - 16x_{53}x_{54} + 4x_{54} - 4x_{54}x_{54} + x_{55}$$

$$R_{41} = -10 + x_9 - x_9x_9 + 8x_9x_{10} - 8x_9x_{23} - 4x_9x_{55} - 16x_{10}x_{10} + 32x_{10}x_{23} + 16x_{10}x_{55} - 16x_{23}x_{23} - 16x_{23}x_{55} - 12x_{48} - 4x_{48}x_{48} - 4x_{55}x_{55} + x_{56}$$

$$R_{42} = 21 + 24x_{48} - 8x_{48}x_{48} - 16x_{48}x_{56} - 2x_{56} - 4x_{56}x_{56} + x_{57}$$

$$R_{43} = 1 - 8x_8 - 4x_8x_{10} + 8x_8x_{59} - 16x_9 - 8x_9x_{10} + 16x_9x_{59} + 2x_{10} + 4x_{10}x_{19} - 4x_{10}x_{57} + 8x_{19} - 8x_{19}x_{59} - 4x_{57}x_{57} + x_{58} + 3x_{59} - 4x_{59}x_{59}$$

$$R_{44} = -6x_8 - 3x_8x_8 - 18x_8x_9 - 16x_9 - 27x_9x_9 + 4x_9x_{10} + 3x_{10} + 3x_{10}x_{10} - 4x_{10}x_{19} - 19x_{19} - 8x_{19}x_{19} + x_{20} + x_{59} - 4x_{71}$$

$$R_{45} = -28x_8 - 24x_8x_{10} + 16x_8x_{58} + 16x_8x_{59} - 56x_9 - 48x_9x_{10} + 32x_9x_{58} + 32x_9x_{59} + 6x_{10} + 6x_{10}x_{10} + 24x_{10}x_{19} + 28x_{19} - 16x_{19}x_{58} - 16x_{19}x_{59} - 4x_{58}x_{58} + 2x_{59} - 8x_{59}x_{59} + x_{60}$$

$$R_{46} = -2 - 4x_{60} - 4x_{60}x_{60} + x_{61}$$

$$R_{47} = -x_3x_{13} + x_{62}$$

$$R_{48} = x_{12} - x_{13} - 2x_{22}x_{62} - x_{63}$$

$$R_{49} = x_{12} - x_{13} - 2x_{26}x_{62} - x_{64}$$

$$R_{50} = -2x_{28}x_{62} + 2x_{41}x_{62} - 2x_{62} - x_{65} + x_{70}$$

$$R_{51} = x_{12} - x_{13} - 2x_{30}x_{62} - x_{66}$$

$$R_{52} = x_{12} - x_{13} - 2x_{32}x_{62} + 6x_{62} - x_{67}$$

$$R_{53} = x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68}$$

$$R_{54} = x_{12} - x_{13} - 2x_{38}x_{62} - x_{69}$$

$$R_{55} = x_{12} - x_{13} - 2x_{41}x_{62} - 4x_{62} - x_{70}$$

$$R_{56} = -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}x_{10} + x_{71}$$

$$R_{57} = 1 + 2x_{11} + x_{11}x_{11} - x_{71}x_{71} + x_{72}$$

$$R_{58} = -4x_{60}x_{62} - 2x_{61}x_{62} + 2x_{62} + x_{73}$$

$$R_{59} = -x_{12}x_{12} + 2x_{12}x_{13} + 2x_{12}x_{73} - x_{13}x_{13} - 2x_{13}x_{73} - x_{72}x_{72} - x_{73}x_{73} + x_{74}$$

$$R_{60} = -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77}$$

$$R_{61} = -1 + 2x_3 + 2x_3x_8 + 4x_3x_9 - 2x_3x_{19} + x_{75}$$

$$R_{62} = -x_8x_{214} - x_{12} + x_{76} + x_{214}$$

$$R_{63} = -2x_8x_{62} - 4x_9x_{62} + x_9x_{214} + x_{12} - x_{13} - 2x_{19}x_{62} + x_{78}$$

$$R_{64} = -x_{12}x_{63} + x_{13}x_{63} + x_{79}$$

$$R_{65} = -x_{64}x_{79} + x_{80}$$

$$R_{66} = -x_{65}x_{80} + x_{81}$$

$$R_{67} = -x_{66}x_{81} + x_{82}$$

$$R_{68} = -x_{67}x_{82} + x_{83}$$

$$R_{69} = -x_{68}x_{83} + x_{84}$$

$$R_{70} = -x_{69}x_{84} + x_{85}$$

$$R_{71} = -x_{70}x_{85} + x_{86}$$

$$R_{72} = -x_{77}x_{86} + x_{87}$$

$$R_{73} = -x_{78}x_{87} + x_{88}$$

$$R_{74} = -x_{88}x_{88} + x_{89}$$

$$R_{75} = -x_2 - x_2x_{16} + x_{16} + x_{90}$$

$$R_{76} = -x_{14}x_{90} + x_{91}$$

$$\begin{aligned} R_{77} = & -2x_1x_1 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} \\ & - x_{12}x_{12} - 2x_{12}x_{91} - x_{13}x_{13} - 2x_{13}x_{92} + x_{17}x_{93} \\ & - x_{74}x_{89} - x_{91}x_{91} - x_{92}x_{92} \end{aligned}$$

$$R_{78} = -x_2x_{15} + x_{15} - x_{15}x_{90} + x_{92}$$

$$R_{79} = -x_{96} - 4x_{96}x_{208} + x_{98}$$

$$R_{80} = -x_4x_{119} + x_{94}x_{200} - x_{97} - x_{98}x_{119}$$

$$R_{81} = x_{99}x_{200} - x_{100}x_{122} - x_{101}$$

$$R_{82} = x_{102}x_{200} - x_{103}x_{122} - x_{104}$$

$$R_{83} = x_{105}x_{200} - x_{106}x_{122} - x_{107}$$

$$R_{84} = x_{108}x_{200} - x_{109}x_{122} - x_{110}$$

$$R_{85} = x_{111}x_{201} - x_{113}x_{122} - x_{114}$$

$$R_{86} = x_{115}x_{201} - x_{116}x_{122} - x_{117}$$

$$R_{87} = -x_{17}x_{119} + x_{122}$$

$$R_{88} = x_{118} - x_{120}x_{122} - x_{121}$$

$$R_{89} = -x_{122}x_{124} + x_{123} - x_{125}$$

$$R_{90} = -x_{122}x_{127} + x_{126}x_{200} - x_{128}$$

$$R_{91} = -x_{122}x_{130} + x_{129}x_{200} - x_{131}$$

$$R_{92} = 4x_{132} - 8x_{132}x_{134} + x_{135}$$

$$R_{93} = -4x_{132}x_{135} - x_{135} + x_{136}$$

$$R_{94} = 2x_{133} + x_{133}x_{133} + x_{135}x_{135} - x_{136}x_{136}$$

$$R_{95} = -x_{133}x_{138} - x_{138} + x_{139}$$

$$R_{96} = -1 + x_{137}x_{137} + 2x_{137}x_{139} + x_{138}x_{138}$$

$$\begin{aligned} R_{97} = & 1 + 16x_0 + 64x_0x_0 - 16x_0x_{133} - 4x_{133} \\ & + x_{140} \end{aligned}$$

$$R_{98} = -8x_0x_{138} + x_3 + x_{133}x_{138} - x_{137} - x_{138} - x_{139} + x_{140}x_{141}$$

$$R_{99} = 1 - 2x_4 + x_4x_4 - 2x_4x_{133} + x_{142}$$

$$R_{100} = -x_4x_{174} + x_{17} + x_{142}x_{143} - x_{173}$$

$$R_{101} = -1 + 2x_{133}x_{185} - x_{144} + 2x_{185} - x_{185}x_{185}$$

$$R_{102} = 1 + x_{144}x_{145} - x_{173} - x_{174}x_{185} + 4x_{208}$$

$$R_{103} = -x_{133}x_{147} - x_{147} + x_{148}$$

$$R_{104} = -1 + x_{146}x_{146} + 2x_{146}x_{148} + x_{147}x_{147}$$

$$R_{105} = 1 + x_{133} - x_{133}x_{150} - x_{150} + x_{151}$$

$$R_{106} = x_{149}x_{149} + 2x_{149}x_{151} - 2x_{150} + x_{150}x_{150}$$

$$R_{107} = -x_{133}x_{153} - x_{153} + x_{154}$$

$$R_{108} = -1 + x_{152}x_{152} + 2x_{152}x_{154} + x_{153}x_{153}$$

$$R_{109} = -x_{133}x_{156} - x_{156} + x_{157}$$

$$R_{110} = -1 + x_{155}x_{155} + 2x_{155}x_{157} + x_{156}x_{156}$$

$$R_{111} = -x_{133}x_{159} - x_{159} + x_{160}$$

$$R_{112} = -1 + x_{158}x_{158} + 2x_{158}x_{160} + x_{159}x_{159}$$

$$R_{113} = -x_{133}x_{162} - x_{162} + x_{163}$$

$$R_{114} = -1 + x_{161}x_{161} + 2x_{161}x_{163} + x_{162}x_{162}$$

$$R_{115} = -x_{133}x_{165} - x_{165} + x_{166}$$

$$R_{116} = -1 + x_{164}x_{164} + 2x_{164}x_{166} + x_{165}x_{165}$$

$$R_{117} = -x_{133}x_{168} - x_{168} + x_{169}$$

$$R_{118} = -1 + x_{167}x_{167} + 2x_{167}x_{169} + x_{168}x_{168}$$

$$R_{119} = -x_{133}x_{171} - x_{171} + x_{172}$$

$$R_{120} = -1 + x_{170}x_{170} + 2x_{170}x_{172} + x_{171}x_{171}$$

$$R_{121} = 1 - x_{133}x_{174} - x_{174} + x_{175}$$

$$R_{122} = -1 + 2x_{173} + x_{173}x_{173} + 2x_{173}x_{175} + x_{174}x_{174}$$

$$R_{123} = -x_{133}x_{177} - x_{177} + x_{178}$$

$$R_{124} = -1 + x_{176}x_{176} + 2x_{176}x_{178} + x_{177}x_{177}$$

$$R_{125} = -x_{133}x_{180} - x_{180} + x_{181}$$

$$R_{126} = -1 + x_{179}x_{179} + 2x_{179}x_{181} + x_{180}x_{180}$$

$$R_{127} = -x_{133}x_{183} - x_{183} + x_{184}$$

$$R_{128} = -1 + x_{182}x_{182} + 2x_{182}x_{184} + x_{183}x_{183}$$

$$\begin{aligned} R_{129} = & -22 - x_0x_0 - x_1x_1 - x_2x_2 - x_3x_3 \\ & - x_4x_4 - x_8x_8 - x_9x_9 - x_{10}x_{10} - x_{11}x_{11} \\ & - x_{12}x_{12} - x_{13}x_{13} - x_{14}x_{14} - x_{15}x_{15} - x_{16}x_{16} \\ & - x_{17}x_{17} - x_{19}x_{19} - x_{22}x_{22} - x_{26}x_{26} - x_{28}x_{28} \\ & - x_{30}x_{30} - x_{32}x_{32} - x_{34}x_{34} - x_{35}x_{35} - x_{38}x_{38} \\ & - x_{41}x_{41} - x_{60}x_{60} - x_{61}x_{61} - x_{94}x_{94} - x_{95}x_{95} \\ & - x_{99}x_{99} - x_{102}x_{102} - x_{105}x_{105} - x_{108}x_{108} - x_{111}x_{111} \\ & - x_{112}x_{112} - x_{115}x_{115} - x_{118}x_{118} - x_{119}x_{119} - x_{123}x_{123} \\ & - x_{126}x_{126} - x_{129}x_{129} + x_{132} - x_{185}x_{185} - x_{208}x_{208} \end{aligned}$$

$$\begin{aligned} R_{130} = & -x_4 - x_4x_8 - x_4x_9 - x_4x_{10} - x_4x_{11} \\ & - x_4x_{12} - x_4x_{13} - x_4x_{14} - x_4x_{15} - x_4x_{16} \\ & + x_{185} \end{aligned}$$

$$R_{131} = -x_{133}x_{187} - x_{187} + x_{196}$$

$$R_{132} = -1 + x_{186}x_{186} + x_{187}x_{187} - x_{196}x_{196}$$

$$R_{133} = -x_{189} - x_{189}x_{190} + x_{197}$$

$$R_{134} = -1 + x_{188}x_{188} + x_{189}x_{189} - x_{197}x_{197}$$

$$R_{135} = -x_{183}x_{183} + x_{198}$$

$$R_{136} = x_{187} - 2x_{192}x_{198} - x_{198}$$

$$R_{137} = -4x_{187} - 8x_{187}x_{191} + x_{190}$$

$$R_{138} = -x_{133} - x_{186}x_{193} + x_{190}$$

$$R_{139} = -x_{182} - x_{184} - x_{186}x_{194} + x_{188}$$

$$\begin{aligned} R_{140} = & -x_8 - x_9 - 4x_{19} - 2x_{22} - 2x_{26} \\ & - 2x_{28} - 2x_{30} - 2x_{32} - 4x_{34} - 2x_{35} \\ & - 2x_{38} - 2x_{41} - 4x_{60} - 2x_{61} - 4x_{183}x_{195} \\ & + x_{189} \end{aligned}$$

$$\begin{aligned} R_{141} = & 24 + 64x_0x_0 - 16x_0x_{138} + x_1x_1 - 2x_1x_{174} \\ & + x_2x_2 - 2x_2x_{165} + x_3x_3 + 2x_3x_{138} - 2x_3x_{147} \\ & + x_8x_8 - 2x_8x_{150} - 2x_8x_{183} + x_9x_9 - 2x_9x_{153} \\ & - 2x_9x_{183} + x_{10}x_{10} - 2x_{10}x_{156} + x_{11}x_{11} - 2x_{11}x_{159} \\ & + x_{12}x_{12} - 2x_{12}x_{162} + x_{13}x_{13} - 2x_{13}x_{168} + x_{14}x_{14} \\ & - 2x_{14}x_{171} + x_{15}x_{15} - 2x_{15}x_{177} + x_{16}x_{16} - 2x_{16}x_{180} \\ & - 8x_{19}x_{183} - 4x_{22}x_{183} - 4x_{26}x_{183} - 4x_{28}x_{183} - 4x_{30}x_{183} \\ & - 4x_{32}x_{183} - 8x_{34}x_{183} - 4x_{35}x_{183} - 4x_{38}x_{183} - 4x_{41}x_{183} \\ & - 8x_{60}x_{183} - 4x_{61}x_{183} - x_{133}x_{199} - 2x_{137} - 2x_{138}x_{147} \end{aligned}$$

$$
\begin{aligned}
&- 2x_{139} + x_{139}x_{139} - 2x_{146} - x_{147}x_{147} - 2x_{148} \\
&+ x_{148}x_{148} - 2x_{149} + 4x_{150} - x_{150}x_{150} - 2x_{151} \\
&+ x_{151}x_{151} - 2x_{152} - x_{153}x_{153} - 2x_{154} + x_{154}x_{154} \\
&- 2x_{155} - x_{156}x_{156} - 2x_{157} + x_{157}x_{157} - 2x_{158} \\
&- x_{159}x_{159} - 2x_{160} + x_{160}x_{160} - 2x_{161} - x_{162}x_{162} \\
&- 2x_{163} + x_{163}x_{163} - 2x_{164} - x_{165}x_{165} - 2x_{166} \\
&+ x_{166}x_{166} - 2x_{167} - x_{168}x_{168} - 2x_{169} + x_{169}x_{169} \\
&- 2x_{170} - x_{171}x_{171} - 2x_{172} + x_{172}x_{172} - 2x_{173} \\
&- x_{174}x_{174} + x_{175}x_{175} - 2x_{176} - x_{177}x_{177} - 2x_{178} \\
&+ x_{178}x_{178} - 2x_{179} - x_{180}x_{180} - 2x_{181} + x_{181}x_{181} \\
&- 2x_{182} + x_{183}x_{183} - 2x_{184} + x_{184}x_{184} - x_{202}x_{202} \\
&- x_{203}x_{203} - x_{204} - x_{204}x_{204}
\end{aligned}
$$

$$
R_{142} = - x_{95}x_{200} + x_{119}
$$

$$
R_{143} = - x_{112}x_{201} + x_{119}
$$

$$
R_{144} = - 1 + x_2 - x_3 - 2x_3x_{209}
$$

$$
\begin{aligned}
R_{145} = &- 1 - 8x_8x_{19} - 4x_8x_{22} - 4x_8x_{26} - 4x_8x_{28} \\
&- 4x_8x_{30} - 4x_8x_{32} - 8x_8x_{34} - 4x_8x_{35} - 4x_8x_{38} \\
&- 4x_8x_{41} - 8x_8x_{60} - 4x_8x_{61} - 8x_9x_{19} - 4x_9x_{22} \\
&- 4x_9x_{26} - 4x_9x_{28} - 4x_9x_{30} - 4x_9x_{32} - 8x_9x_{34} \\
&- 4x_9x_{35} - 4x_9x_{38} - 4x_9x_{41} - 8x_9x_{60} - 4x_9x_{61} \\
&- x_{19} - 16x_{19}x_{19} - 16x_{19}x_{22} - 16x_{19}x_{26} - 16x_{19}x_{28} \\
&- 16x_{19}x_{30} - 16x_{19}x_{32} - 32x_{19}x_{34} - 16x_{19}x_{35} - 16x_{19}x_{38} \\
&- 16x_{19}x_{41} - 32x_{19}x_{60} - 16x_{19}x_{61} - 4x_{22}x_{22} - 8x_{22}x_{26} \\
&- 8x_{22}x_{28} - 8x_{22}x_{30} - 8x_{22}x_{32} - 16x_{22}x_{34} - 8x_{22}x_{35} \\
&- 8x_{22}x_{38} - 8x_{22}x_{41} - 16x_{22}x_{60} - 8x_{22}x_{61} - 4x_{26}x_{26} \\
&- 8x_{26}x_{28} - 8x_{26}x_{30} - 8x_{26}x_{32} - 16x_{26}x_{34} - 8x_{26}x_{35} \\
&- 8x_{26}x_{38} - 8x_{26}x_{41} - 16x_{26}x_{60} - 8x_{26}x_{61} - 4x_{28}x_{28} \\
&- 8x_{28}x_{30} - 8x_{28}x_{32} - 16x_{28}x_{34} - 8x_{28}x_{35} - 8x_{28}x_{38} \\
&- 8x_{28}x_{41} - 16x_{28}x_{60} - 8x_{28}x_{61} - 4x_{30}x_{30} - 8x_{30}x_{32} \\
&- 16x_{30}x_{34} - 8x_{30}x_{35} - 8x_{30}x_{38} - 8x_{30}x_{41} - 16x_{30}x_{60} \\
&- 8x_{30}x_{61} - 4x_{32}x_{32} - 16x_{32}x_{34} - 8x_{32}x_{35} - 8x_{32}x_{38} \\
&- 8x_{32}x_{41} - 16x_{32}x_{60} - 8x_{32}x_{61} - 16x_{34}x_{34} - 16x_{34}x_{35} \\
&- 16x_{34}x_{38} - 16x_{34}x_{41} - 32x_{34}x_{60} - 16x_{34}x_{61} - 4x_{35}x_{35} \\
&- 8x_{35}x_{38} - 8x_{35}x_{41} - 16x_{35}x_{60} - 8x_{35}x_{61} - 4x_{38}x_{38} \\
&- 8x_{38}x_{41} - 16x_{38}x_{60} - 8x_{38}x_{61} - 4x_{41}x_{41} - 16x_{41}x_{60} \\
&- 8x_{41}x_{61} - 16x_{60}x_{60} - 16x_{60}x_{61} - 4x_{61}x_{61} - x_{138}x_{138} \\
&- x_{147}x_{147} + 2x_{150} - x_{150}x_{150} - x_{153}x_{153} - x_{156}x_{156} \\
&- x_{159}x_{159} - x_{162}x_{162} - x_{165}x_{165} - x_{168}x_{168} - x_{171}x_{171} \\
&- x_{174}x_{174} - x_{177}x_{177} - x_{180}x_{180} + x_{183}x_{183} - x_{202}x_{202} \\
&- x_{203}x_{203} - x_{204} - x_{204}x_{204}
\end{aligned}
$$

$$
\begin{aligned}
R_{146} = &- 1 - 4x_{97}x_{97} - 4x_{101}x_{101} - 4x_{104}x_{104} - 4x_{107}x_{107} \\
&- 4x_{110}x_{110} - 4x_{114}x_{114} - 4x_{117}x_{117} + x_{119}x_{119} - 4x_{121}x_{121} \\
&- 4x_{125}x_{125} - 4x_{128}x_{128} - 4x_{131}x_{131} - x_{205}x_{205} - x_{206}x_{206} \\
&- x_{207} - x_{207}x_{207}
\end{aligned}
$$

$$
\begin{aligned}
R_{147} = &- 2x_{94}x_{147} + 2x_{99} - 2x_{99}x_{150} - 2x_{102}x_{153} - 2x_{105}x_{156} \\
&- 2x_{108}x_{159} - 2x_{111}x_{162} - 2x_{115}x_{168} - 2x_{118}x_{171} - 2x_{123}x_{177} \\
&- 2x_{126}x_{180} - 2x_{129}x_{183} - 2x_{133}x_{211} + 2x_{146}x_{147} + 2x_{147}x_{148}
\end{aligned}
$$

$$
\begin{aligned}
&- 2x_{149} + 2x_{149}x_{150} + 2x_{150}x_{151} - 2x_{151} + 2x_{152}x_{153} \\
&+ 2x_{153}x_{154} + 2x_{155}x_{156} + 2x_{156}x_{157} + 2x_{158}x_{159} + 2x_{159}x_{160} \\
&+ 2x_{161}x_{162} + 2x_{162}x_{163} + 2x_{167}x_{168} + 2x_{168}x_{169} + 2x_{170}x_{171} \\
&+ 2x_{171}x_{172} + 2x_{176}x_{177} + 2x_{177}x_{178} + 2x_{179}x_{180} + 2x_{180}x_{181} \\
&+ 2x_{182}x_{183} + 2x_{183}x_{184} + 4x_{208}x_{211} + x_{210}
\end{aligned}
$$

$$
\begin{aligned}
R_{148} = \; &- 1 + x_{94}x_{94} - 2x_{94}x_{146} + x_{99}x_{99} - 2x_{99}x_{149} \\
&+ x_{102}x_{102} - 2x_{102}x_{152} + x_{105}x_{105} - 2x_{105}x_{155} + x_{108}x_{108} \\
&- 2x_{108}x_{158} + x_{111}x_{111} - 2x_{111}x_{161} + x_{115}x_{115} - 2x_{115}x_{167} \\
&+ x_{118}x_{118} - 2x_{118}x_{170} + x_{123}x_{123} - 2x_{123}x_{176} + x_{126}x_{126} \\
&- 2x_{126}x_{179} + x_{129}x_{129} - 2x_{129}x_{182} + x_{146}x_{146} - x_{147}x_{147} \\
&+ x_{149}x_{149} + 2x_{150} - x_{150}x_{150} + x_{152}x_{152} - x_{153}x_{153} \\
&+ x_{155}x_{155} - x_{156}x_{156} + x_{158}x_{158} - x_{159}x_{159} + x_{161}x_{161} \\
&- x_{162}x_{162} + x_{167}x_{167} - x_{168}x_{168} + x_{170}x_{170} - x_{171}x_{171} \\
&+ x_{176}x_{176} - x_{177}x_{177} + x_{179}x_{179} - x_{180}x_{180} + x_{182}x_{182} \\
&- x_{183}x_{183} - 4x_{208}x_{210} - 2x_{210} + x_{211}
\end{aligned}
$$

$$
\begin{aligned}
R_{149} = \; &- 2x_{95}x_{138} - 2x_{112}x_{165} - 2x_{119}x_{174} - 2x_{133}x_{213} + 2x_{137}x_{138} \\
&+ 2x_{138}x_{139} + 2x_{164}x_{165} + 2x_{165}x_{166} + 2x_{173}x_{174} + 2x_{174} \\
&+ 2x_{174}x_{175} + 4x_{208}x_{213} + x_{212} - x_{213}
\end{aligned}
$$

$$
\begin{aligned}
R_{150} = \; &x_{95}x_{95} - 2x_{95}x_{137} + x_{112}x_{112} - 2x_{112}x_{164} + x_{119}x_{119} \\
&- 2x_{119}x_{173} + x_{137}x_{137} - x_{138}x_{138} + x_{164}x_{164} - x_{165}x_{165} \\
&+ x_{173}x_{173} - x_{174}x_{174} - 4x_{208}x_{212} - x_{212} + x_{213}
\end{aligned}
$$

$$
R_{151} = \; - x_3x_{12} + x_{214}
$$